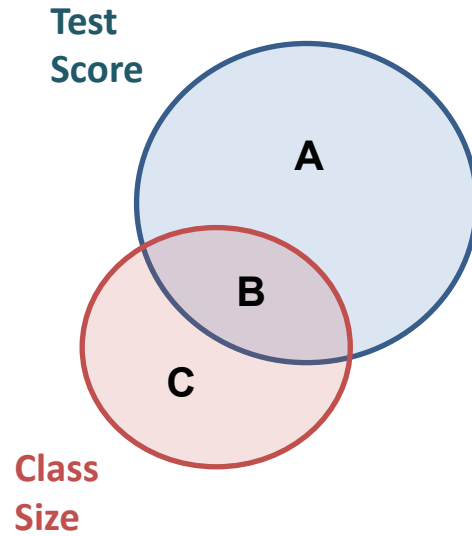
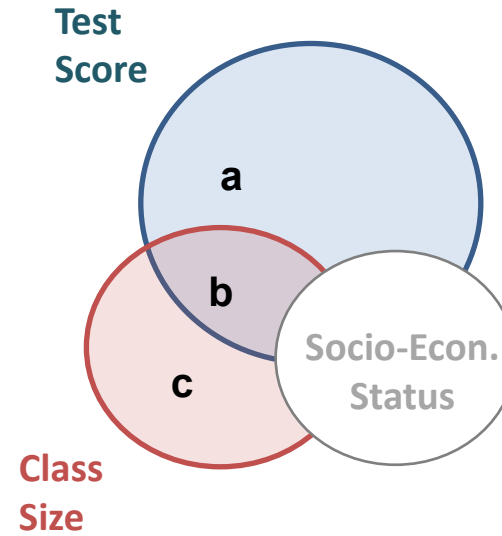


BIAS FROM SELECTION AND CORRECTIONS USING MATCHING



$$\text{slope} : \frac{B}{B+C} \rightarrow \frac{b}{b+c}$$

$$SE_{b1} : \frac{A}{B+C} \rightarrow \frac{a}{b+c}$$



When we add a control variable that IS correlated with the policy variable it affects both the slope and the standard error.

“There are two types of people”

Microfinance example of bias from selection

Number of people per category

	Not an Entrepreneur	Entrepreneur
No Loan	30	15
Takes a Loan	20	35

Average weekly income per category

	Not an Entrepreneur	Entrepreneur
No Loan	\$10	\$20
Takes a Loan	\$10	\$20

$$\text{No Microfinance Loan: } \frac{30 \cdot \$10 + 15 \cdot \$20}{45} = \$13.33$$

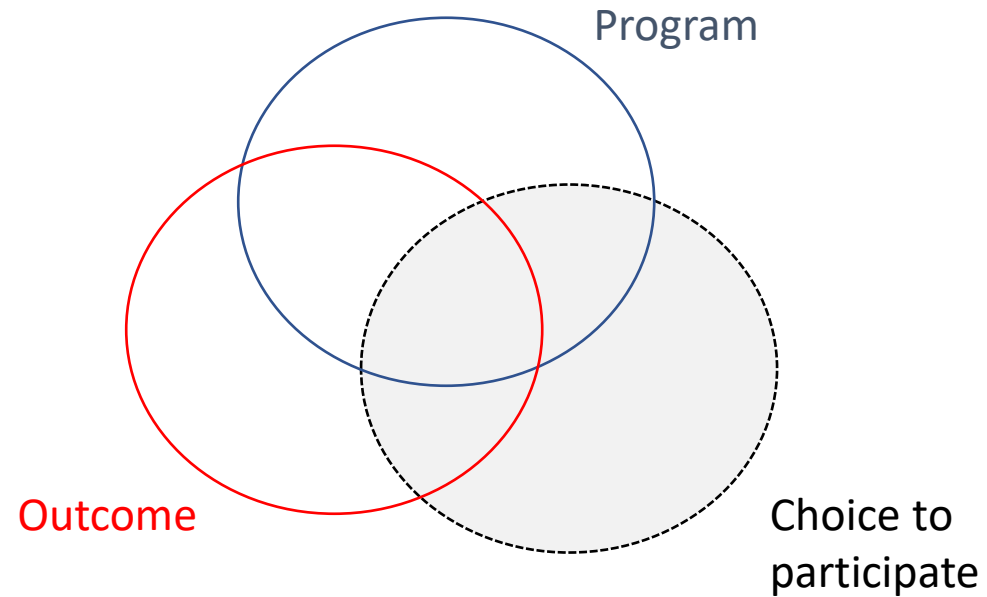
$$\text{Yes Microfinance Loan: } \frac{20 \cdot \$10 + 35 \cdot \$20}{55} = \$16.37$$

The loan appears to have an impact even when we know it didn't!

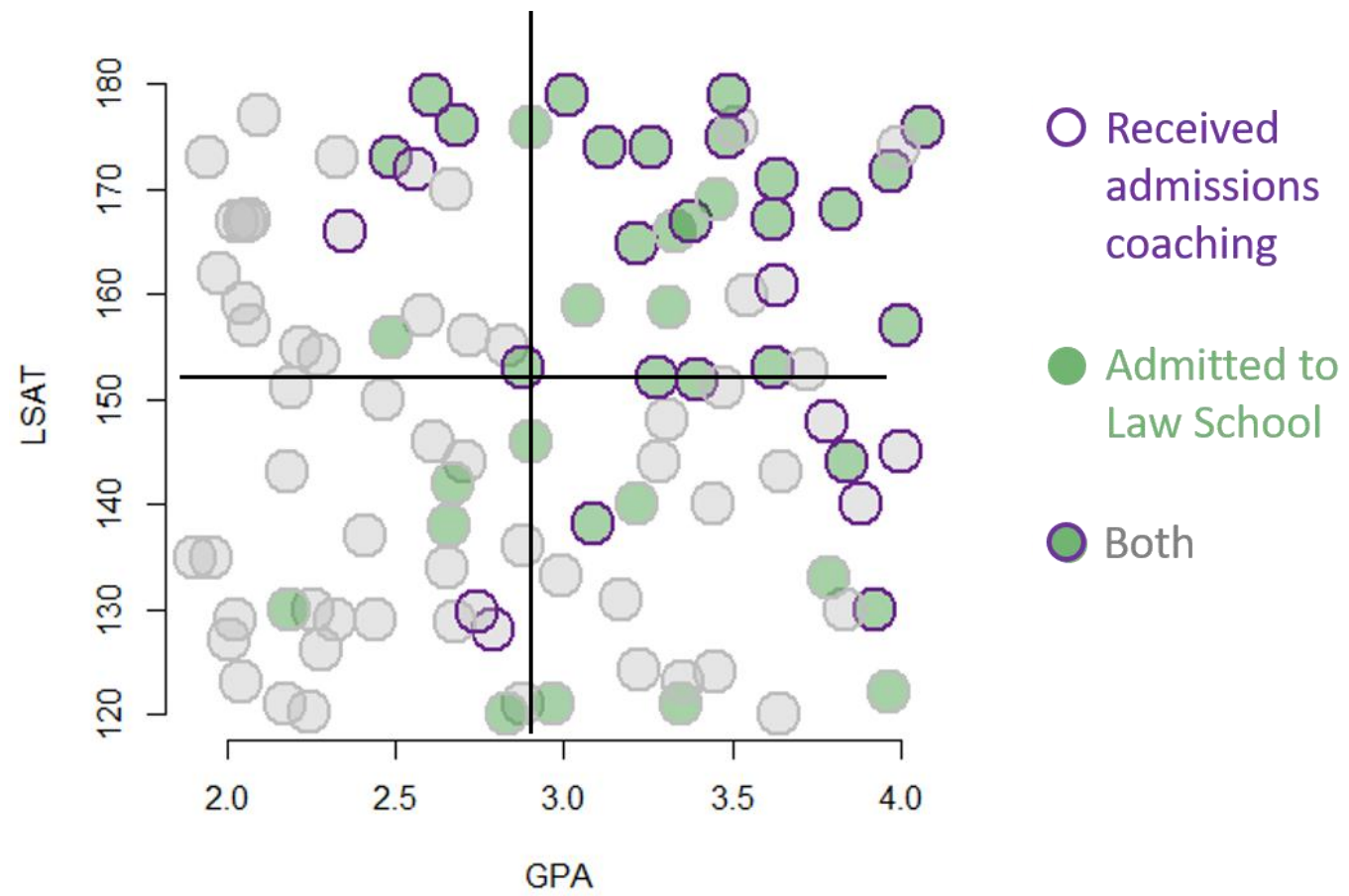
Charter school example

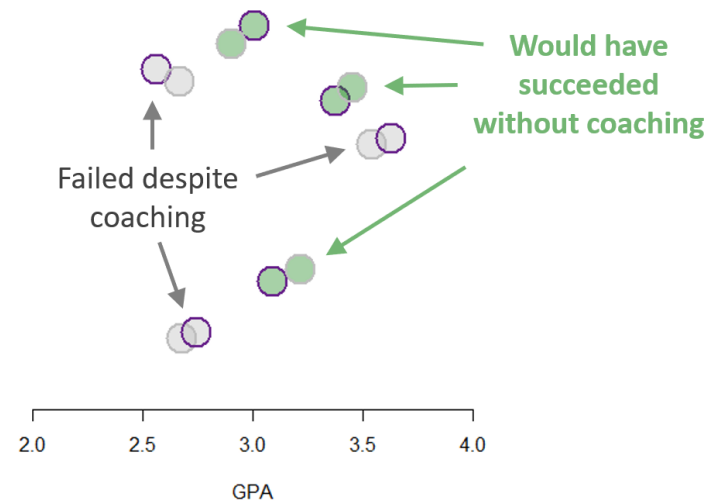
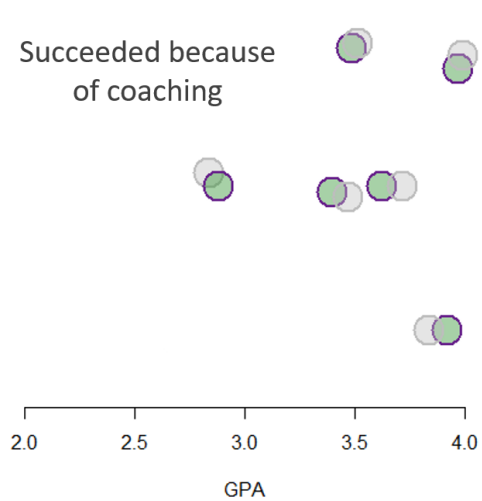
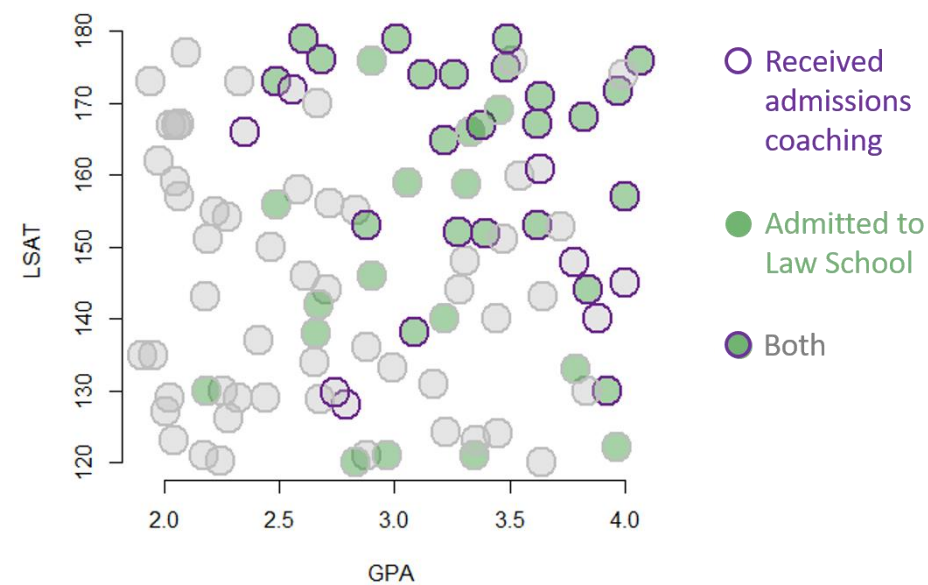
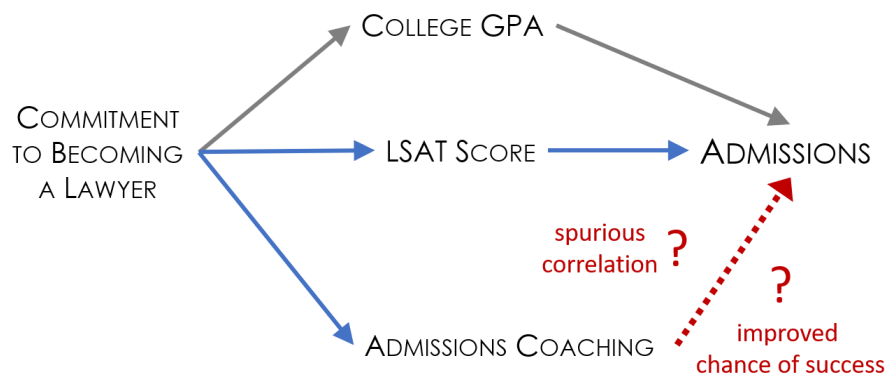
		GRADE 4				GRADE 8		
		2003	2005	2007	2009	2005	2007	2009
Overall		▲	▲	▲	▲	▲	▲	▲
Free or reduced-price lunch								
	Eligible	▲	◆	▲	▲	◆	◆	◆
	Not eligible	▲	▲	◆	▲	▲	▲	▲
City Location		▲	◆	▲	▲	◆	◆	▲
Race/ethnicity								
	White	◆	◆	◆	◆	◆	◆	◆
	Black	◆	◆	◆	◆	▲	◆	◆
	Hispanic	◆	◆	◆	◆	◆	◆	◆
▲ Other public schools score significantly higher than charter schools								
◆ Other public schools not significantly different than charter schools								
▼ Other public schools score significantly lower than charter schools								

The “selection” problem

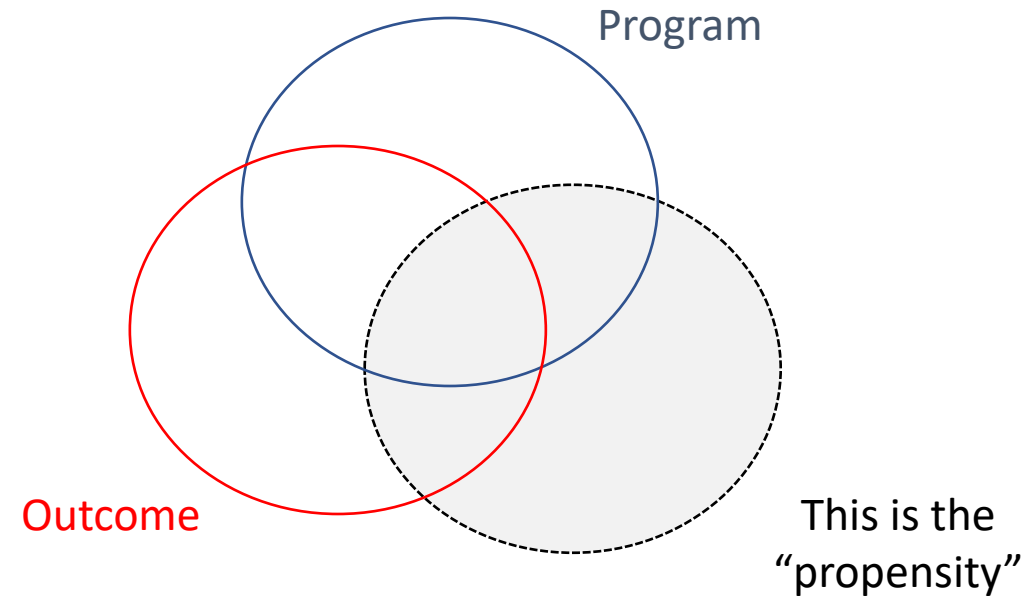


When happens when people are given the choice to participate in a program? When do problems arise?
How do we estimate program impact in this case?



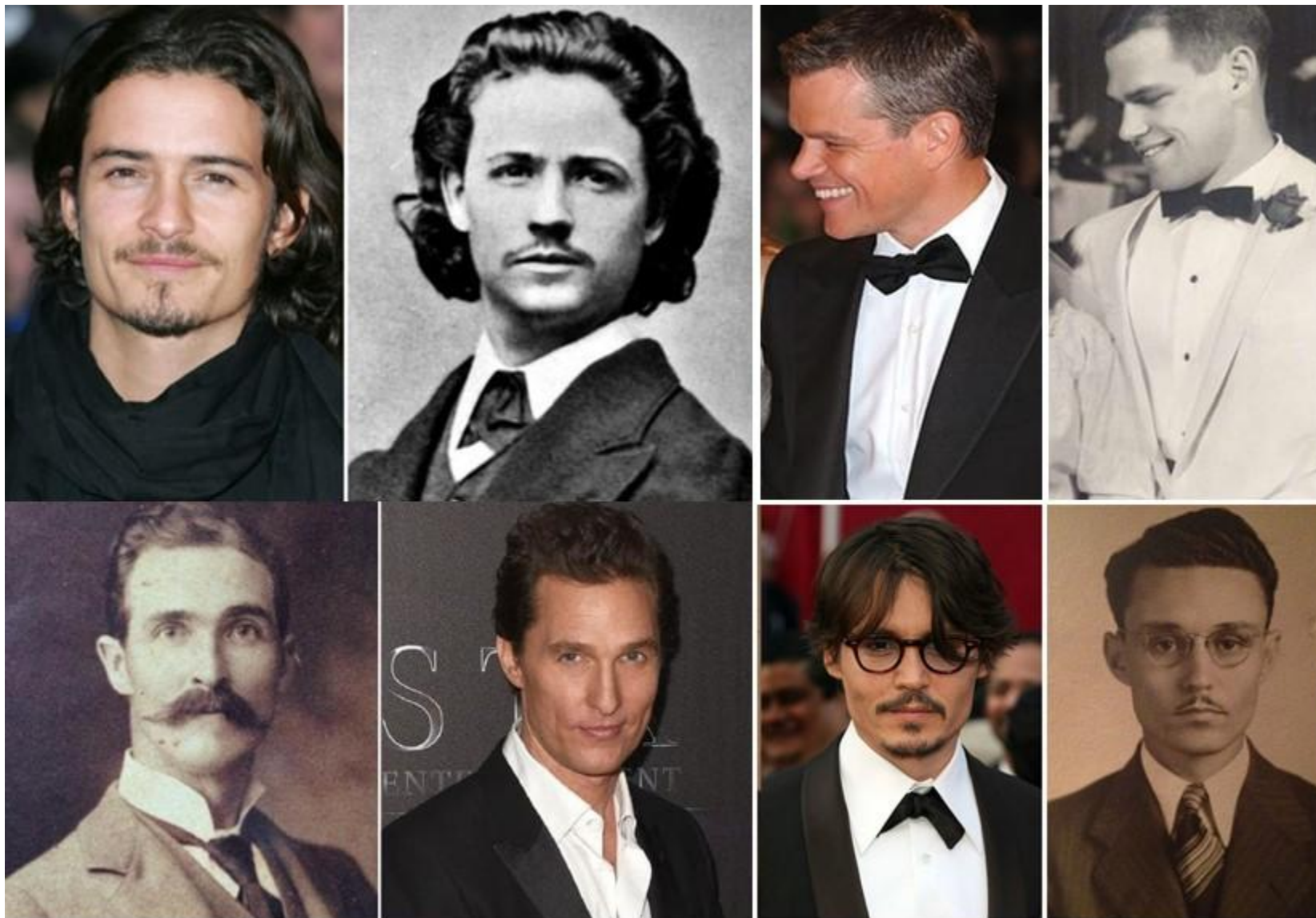


Propensity score matching



Selection bias is just another case of omitted variable bias

Matching

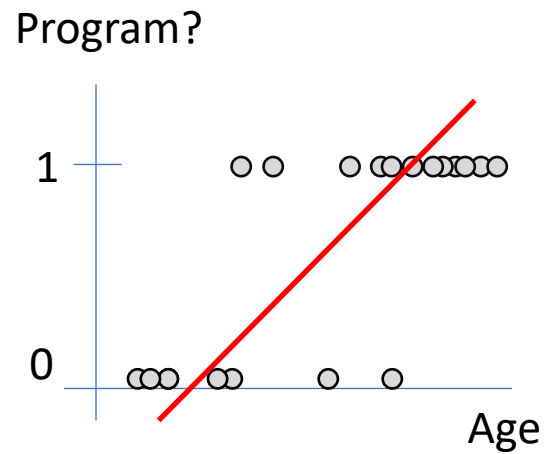


Propensity score matching

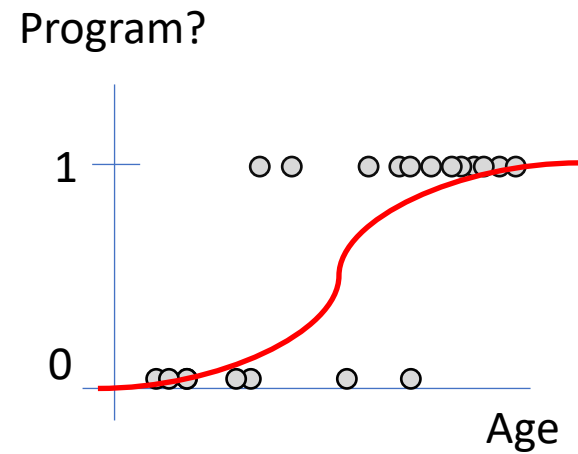
ID#	Age	Group	Outcome	Program?	
#1	40	Blue	\$75	0	
#2	43	Blue	\$64	1	
#3	54	Blue	\$69	0	
#4	23	Red	\$32	0	
#5	41	Red	\$49	0	
#6	38	Red	\$87	1	

Probability Models

**Linear Probability
Model (OLS)**



**Logit
Model**



These models predict the probability of participation in the program, i.e. the “propensity” of an individual

Propensity score matching



ID#	Age	Race	Outcome	Program?	Predicted Propensity
#1	40	Blue	\$75	0	0.87
#2	43	Blue	\$64	1	0.98
#3	54	Blue	\$69	0	0.45
#4	23	Red	\$32	0	0.53
#5	41	Red	\$49	0	1.02
#6	38	Red	\$87	1	1.11

We add a variable for the predicted propensity score.
Remember not to use your outcome in your propensity score calculations.

Propensity score matching

ID#	Age	Race	Outcome	Program?	Predicted Propensity
#1	40	Blue	\$75	0	0.87
#2	43	Blue	\$64	1	0.98
#3	54	Blue	\$69	0	0.45
#4	23	Red	\$32	0	0.53
#5	41	Red	\$49	0	1.02
#6	38	Red	\$87	1	1.11

Then matches are made to form
a synthetic control group:

#5 <--> #6

#2 <--> #1

Nearest Neighbors

id	p.score	books	movies	fiction	scifi	children	male	age
9	0.636	1	1	1	1	0	1	25
10	0.644	1	1	1	0	0	0	25
11	0.64	1	1	1	1	0	0	26
12	0.632	0	1	1	0	0	0	26
13	0.617	1	1	0	1	0	1	24

Harry Potter Example

Propensity Score

1.67169631	No	Yes	Yes	Yes	No	Male	25	Black	<--	
1.70500761	Yes	Yes	Yes	Yes	No	Female	25	Black	<--	Good matches
1.90810452	Yes	Yes	No	Yes	No	Male	37	White	<--	(these are bad matches)
1.94231859	No	No	No	No	No	Male	25	Asian	<--	
1.96650156	No	Yes	No	No	No	Male	34	White	<--	
1.97086420	No	Yes	Yes	Yes	No	Male	37	White	<--	
1.98329369	No	Yes	No	Yes	No	Male	34	White	<--	
2.04605338	Yes	Yes	Yes	Yes	No	Male	34	White	<--	Good matches
2.04605338	No	Yes	Yes	Yes	No	Male	34	White	<--	
2.11269867	No	Yes	Yes	No	No	Female	32	White		
2.19206910	Yes	Yes	No	No	No	Male	25	White	<--	Good matches
2.19643173	No	Yes	Yes	Yes	No	Male	28	White	<--	
2.20018114	No	Yes	Yes	Yes	Yes	Female	35	White	<--	(these are bad matches)
2.20886123	Yes	Yes	No	Yes	No	Male	25	White	<--	
2.21295091	Yes	Yes	Yes	No	No	Female	28	White	<--	Good match
2.23392429	Yes	Yes	No	Yes	No	Male	24	White	<--	OK match
2.26307702	No	Yes	Yes	No	No	Female	26	White	<--	

Harry Potter Example

```
Call:
lm(formula = dv ~ x$age + x$movie + x$fiction + x$fantasy + x$kids +
    x$sex + x$race)

Residuals:
    Min       1Q   Median       3Q      Max
-1.2631 -0.5902  0.1084  0.6867  1.2950

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.60221    0.76328   3.409  0.00147 **
x$age        -0.02506    0.02354  -1.065  0.29317
x$movieYes     0.80016    0.30575   2.617  0.01237 *
x$fictionYes   0.06276    0.23885   0.263  0.79405
x$fantasyYes   0.01679    0.25872   0.065  0.94857
x$kidsYes     0.14588    0.59635   0.245  0.80797
x$sexMale    -0.03331    0.26765  -0.124  0.90156
x$raceBlack  -1.15034    0.42928  -2.680  0.01056 *
x$raceOther  -0.24852    0.65052  -0.382  0.70441
x$raceWhite  -0.55041    0.40219  -1.369  0.17860
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Harry Potter Example

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.712310	1.204111	1.422	0.1928
x\$age	-0.027715	0.030506	-0.909	0.3902
x\$movieYes	-0.019632	0.456722	-0.043	0.9668
x\$fictionYes	0.312937	0.269323	1.162	0.2788
x\$fantasyYes	0.138953	0.321117	0.433	0.6766
x\$kidsYes	-0.009775	0.562699	-0.017	0.9866
x\$sexMale	0.574428	0.467635	1.228	0.2542
x\$raceBlack	-0.845482	0.391054	-2.162	0.0626
x\$raceWhite	-0.920483	0.449148	-2.049	0.0746

propensity

	x.read	m1.fitted.values	
1	Yes	0.460131743	<--
2	No	0.543275854	<--
3	No	-0.122758362	
4	No	0.244294952	
5	Yes	1.111048656	
6	No	0.007672132	
7	Yes	0.856212741	
8	Yes	0.606218309	
9	Yes	0.888951344	
10	Yes	0.792708186	
11	No	0.293273998	
12	No	-0.187643773	
13	No	0.126993201	
14	No	0.404322654	<--
15	No	-0.056651179	
16	No	0.210137311	
17	No	-0.178187769	

1.	Yes	Yes	No	Yes	No	Male	37	White
2.	No	Yes	No	Yes	No	Male	34	White
14.	No	Yes	No	No	No	Male	34	White

How do we know matching has been successful?

TABLE 2

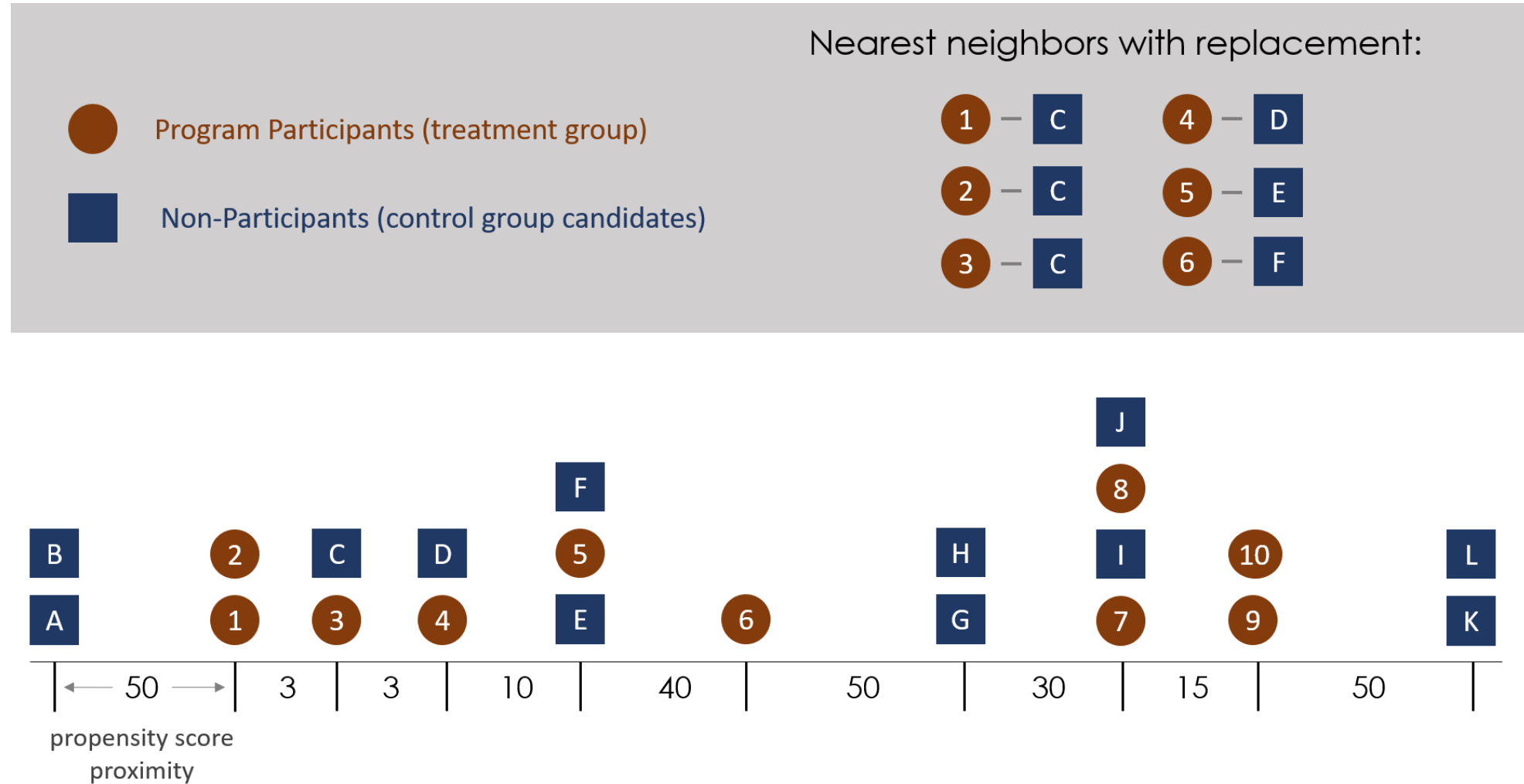
Background Characteristics of Students in Treatment and Control Groups
(Total numbers of cases in parentheses)

Characteristic	All students in the study			All students with scores three or four years after application		
	Choice students	Control students	p value ^a	Choice students	Control students	p value ^a
Math scores before application	39.7 (264)	39.3 (173)	.81	40.0 (61)	40.6 (33)	.86
Reading scores before application	38.9 (266)	39.4 (176)	.74	42.1 (60)	39.2 (33)	.35
Family income	10,860 (423)	12,010 (127)	.14	10,850 (143)	11,170 (25)	.84
Mothers' education 3 = some college 4 = college degree	4.2 (423)	3.9 (127)	.04	4.1 (144)	3.8 (29)	.15
Percent married parents	24 (424)	30 (132)	.17	23 (145)	38 (29)	.11
Parents' time with children 1 = 1–2 hours/week 2 = 3–4 hours/week 3 = 5 or more	1.9 (420)	1.8 (130)	.37	1.9 (140)	1.7 (27)	.26
Parents' education expectations of children 4 = college 5 = graduate school	4.2 (422)	4.2 (129)	.85	4.2 (142)	3.7 (27)	.01

a. The tests of significance are suggestive of the equivalence of the two groups. Technically, tests of significance should be done at each point of random assignment, but the number of cases at each point is too few for such tests to be meaningful.

“balance”

Greedy Search



```
# nearest neighbor is the default method
```

```
m.out <- matchit( books ~ movies + fiction + scifi + children +  
                  male + age + white + black + asian, data=dat,  
                  method = "nearest" )
```

```
# view match statistics
```

```
m.out
```

```
##
```

```
## Call:
```

```
## matchit(formula = books ~ movies + fiction + scifi + children +
```

```
##      male + age + white + black + asian, data = dat, method = "nearest")
```

```
##
```

```
## Sample sizes:
```

```
##           Control Treated
```

```
## All           32      19
```

```
## Matched       19      19
```

```
## Unmatched     13       0
```

```
## Discarded      0       0
```

http://www.youtube.com/watch?v=pNn_oJwZ2sg&feature=related

What are the problems with Propensity Scores?

<http://www.youtube.com/watch?v=Zg7gxAG8YS4&feature=related>

What are the problems with Propensity Scores?



You can only match on observables!

<http://www.youtube.com/watch?v=Zg7gxAG8YS4&feature=related>

COMMON SUPPORT

